

# Dhruv Agrawal

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## Education

**Manipal Institute of Technology, Manipal, KA**

B.Tech, Computer and Communication Engineering

Expected 2028

CGPA: 8.12

## Relevant Coursework

Data Structures & Algorithms | Database Systems (DBMS) | Design & Analysis of Algorithms | Operating Systems | Object-Oriented Software Development | Probability & Optimization | Data Analytics | Discrete Mathematical Structures

## Experience

### Team Robo Manipal

02/2025 – Present

*AI Researcher*

- Directed development of an AI perception and prescriptive decision system integrating instance segmentation, 3D spatial tracking, and multi-sensor fusion for plant-level risk prediction and yield optimization.
- Engineered a deterministic firmware migration framework automating cross-architecture microcontroller porting via semantic abstraction and simulation validation.
- Designed and executed structured research training for 50+ members across ML and Edge AI domains.

### VersionTwo (V2)

12/2025 – Present

*Co-Founder*

(Part-time)

- Built a decentralized AI agent reputation protocol that assigns, validates, and decays trust scores using multi-layer verification and on-chain settlement.
- Co-authored a research paper formalizing the protocol's self-governance dynamics (see Publications), validating emergent cooperation across a 500-agent economic simulation.

## Projects

### Cross-Asset Portfolio Optimization System | Python, pandas | [Github](#)

- Developed a walk-forward portfolio optimization engine using Hierarchical Risk Parity with regime switching across multiple asset classes.
- Delivered **148% return vs. a 97.5% benchmark**, improving portfolio Sharpe ratio from 0.41 to 0.68.

### Autonomous Agricultural Scouting Robot (FarmBot) — Team RoboManipal | [Github](#)

- Built an AI-driven agricultural scouting robot using YOLOv11-Seg for plant-level disease detection and ripeness estimation under dense foliage occlusion.
- Integrated a multi-sensor perception stack (stereo vision, IMU, GPS, soil NPK sensing) on Jetson Orin Nano and Raspberry Pi Pico with a real-time embedded inference pipeline; validated via field testing.

## Achievements

- Podium Finish — Quantitative Finance Competition, IISc Bangalore:** Built a market-neutral FX statistical arbitrage strategy (14-year, 5-second frequency dataset) achieving **Net Sharpe 1.99**, profitable in 12/14 years.
- Technoxian World Robotics Championship 2024 & RoboCup@Work (Germany)** — Represented Team RoboManipal at both international robotics competitions.

## Publications & Patents

- V. Agarwal, S. BS, D. Agrawal, *Emergent Ethical Alignment in Reputation-Based Economies: Karma as a Self-Governing Framework for Heterogeneous Human-AI-Robot Systems* (preprint) — 500-agent simulation showing self-organizing ethical alignment; identified a 12% misalignment collapse threshold ( $R^2 > 0.999$  predictive accuracy).
- D. Agrawal, V. Agarwal, V. Sarda, *Enhanced Bitcoin Crash Prediction Framework: A Unified Hybrid Approach Integrating Econometric Volatility, Machine Learning, and Deep Sequence Modeling* (preprint) — Hybrid EGARCH+LPPL+Attention-LSTM model (10-year BTC data) achieving **ROC AUC 0.9756**; 26-day early warning ahead of the Nov. 2021 crash.
- Patent Filed: A System and a Method for Automatic Cross Architecture Firmware Porting — App. No. 202641069479 (+4 more patents filed)

## Technical Skills

**Core CS:** Data Structures & Algorithms (DSA), Database Management Systems (DBMS), Machine Learning (ML), Operating Systems

**Programming & Software Engineering:** Python, C/C++, Java, JS/CSS, SQL, PostgreSQL, REST APIs, Agile/SDLC, Testing, Git

**Quant & ML:** PyTorch, TensorFlow, EGARCH, LSTM, Attention Mechanisms, Time-series Modeling, Factor Decomposition

## Positions of Responsibility

### Finova, Fintech Club of MIT Manipal

05/2026 – Present

*Vice President (2026-27)*

- Delivered a fintech workshop on AI-agent pipelines for stock market prediction (100+ registrations).
- Selected to lead day-to-day organizational operations for the 2026–27 tenure.

### E-Cell, MIT Manipal

09/2025 – Present

*Executive*

- Orchestrated execution of Manipal Entrepreneurship Summit 2026 (30,000+ attendees).
- Collaborated with 12 E-Cells nationwide to draft the Bharat Yuva Innovation Policy Recommendations.

**Certifications:** McKinsey.org Forward Program — McKinsey.org, Jun 2026